

Siddharth Saminathan

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SKILLS

Programming & Data

Python · SQL

Backend & Systems

FastAPI · Redis (caching, queues) · PostgreSQL · REST APIs · async pipelines · WebSockets · SSE streaming

AI / ML Systems

LLM APIs · RAG pipelines · embeddings · prompt design · multi-model routing · multi-agent orchestration · memory systems

Infra & Deployment

Docker · Fly.io · AWS (basics) · logging · observability · performance optimization

Data

ETL pipelines · Power BI · Tableau · data validation · pipeline optimization

EDUCATION

M.Sc. Statistics and Machine Learning, Linköping University 2020-2022

Sweden Built an ML pipeline for colorectal cancer metastasis prediction from high-dimensional RNA sequencing data (~64,000 features, under 50 samples), delivering a proof-of-concept for clinical decision support. Applied autoencoders for feature compression and representation learning, improving model efficiency on sparse biomedical data. Used PCA for dimensionality reduction and feature selection, streamlining the feature space by over 99%. Performed gene set enrichment analysis and statistical modelling to identify biologically relevant patterns, enabling interpretable insights for oncologists.

B.Tech Computer Science & Engineering, SRM University 2016-2020 India

PROFILE

AI Product and Systems Engineer with 4+ years of experience building and shipping production AI systems across data infrastructure, enterprise workflows, and consumer applications. Focus on taking systems from concept to real users, improving them through measurable outcomes such as latency, cost, and user retention. Experience spans orchestration, real-time pipelines, and memory-driven AI systems, with a strong emphasis on reliability and iteration in production environments.

EXPERIENCE

AI Engineer

Omnex Systems (AquaPro AI), Chennai, India
2025 - Present

- Built a production AI system enabling AI-assisted quality and control workflows (DFMEA, PFMEA, Control Plans, Process Flows, 5W2H, 5 Whys, 8D), improving consistency and reducing manual analysis effort for engineering teams.
- Designed an orchestration layer using tool-calling: selects tools, routes data across internal services, and executes multi-step workflows end-to-end (generate, validate, update).
- Enabled the system to understand production hierarchies and categories, map inputs to the right structures, and suggest updates based on existing data.
- Implemented feedback-aware flows: retrieves similar past concerns, links them, updates records, and suggests impactful actions to users.
- Reduced response latency from ~12.5s to ~8s by restructuring the pipeline and consolidating model calls.
- Improved output consistency and reduced failures with validation layers, retry handling, and fallback logic.
- Implemented observability (structured logging, tracing, and metrics) to monitor system behavior and debug production issues.

COMPETENCIES

Python SQL FastAPI
Redis (caching, queues)
PostgreSQL REST APIs
async pipelines WebSockets
SSE streaming LLM APIs
RAG pipelines embeddings
prompt design multi-model routing
multi-agent orchestration
memory systems Docker Fly.io
AWS (basics) logging
observability
performance optimization
ETL pipelines Power BI Tableau
data validation
pipeline optimization

Co-Founder, AI Product Engineer

Emote (emotenow.app)
2024 - Present

- Built a production AI companion system for hyper-personalized emotional support, users feel remembered and do not start over each session.
- Designed a memory system that tracks emotional context, relationships, and cause-effect patterns across 100s of conversations, improving relevance per user.
- Built the backend from scratch (FastAPI, Redis, PostgreSQL, hosted on Fly.io) with SSE streaming and WebSocket-based real-time pipelines, caching, and multi-model routing.
- Scaled from 0 to 450+ users, with 40+ power users and ~13-15% weekly cohort retention (pre-PMF).
- Reduced inference cost per conversation from ~\$1.20 to ~\$0.023 through model selection, caching, and prompt optimization.
- Improved latency from ~15s to ~3s through pipeline optimization and streaming.
- Built an internal AI product manager that monitors system behaviour, detects anomalies, and improves observability across the production AI stack.
- Built a Reddit bot to automate user acquisition; reduced outreach time from ~6-8 hours to ~30-40 minutes, enabling 50-100 hyper-personalized DMs per session, increasing activation from ~20% to ~75% and weekly retention from ~5% to ~13%.

Data Engineer

Akkodis AB (Ericsson Client), Sweden
Nov 2022 - Nov 2024

- Owned the redesign of data pipelines powering reporting systems, delivering reliable data to cross-functional stakeholders.
- Built and designed end-to-end ETL lifecycle for dashboards (Power BI, Tableau) used by business stakeholders to drive decisions.
- Led migration of enterprise data platform from Azure to on-prem PostgreSQL cloud environment, ensuring data reliability through quality audits.
- Ensured data consistency and reliability during large-scale migration via rigorous data quality audits.

Data Analyst

Positive Integers Pvt Ltd, Chennai, India
Sep 2019 - May 2020

- Built Tableau dashboards for an NBFC to analyze credit risk and loan repayment behavior, enabling data-driven lending decisions.
- Built real-time Grafana dashboards for ID by SPI Cinemas to track the full order-to-table lifecycle, improving operational visibility.
- Built SQL-based ETL workflows to extract, transform, and load data from operational source systems into reporting layers, ensuring data accuracy and timeliness.

KEY ACHIEVEMENTS

- Built Emote from zero to 450+ users with 40+ power users and 13-15% weekly retention, achieving zero paid acquisition. (<https://emotenow.app/>)
- Led enterprise data platform migration (Azure to PostgreSQL) at Akkodis for Ericsson, improving reporting reliability and delivery.